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1 • CREATE DATABASE Bank_ACID_DB;
2
3 • USE Bank_ACID_DB;
4
5 • CREATE TABLE Customer (
6     Customer_ID INT PRIMARY KEY,
7     Customer_Name VARCHAR(100) NOT NULL,
8     Phone VARCHAR(15),
9     City VARCHAR(50)
10 );
11
12 • CREATE TABLE Account (
13     Account_No INT PRIMARY KEY,
14     Customer_ID INT,
15     Account_Type VARCHAR(20),
16     Balance DECIMAL(12,2),
17     Branch VARCHAR(50),
18     FOREIGN KEY (Customer_ID)
19     REFERENCES Customer(Customer_ID)
20 );
21
22 • CREATE TABLE Bank_Transaction (
23     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
24     Account_No INT,
25     Transaction_Type VARCHAR(20),
26     Amount DECIMAL(12,2),
27     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
28     FOREIGN KEY (Account_No)
29     REFERENCES Account(Account_No)
30 );
31
32 • INSERT INTO Customer
33 (Customer_ID, Customer_Name, Phone, City)
34 VALUES
35 (101, 'Hasini', '9876543210', 'Hyderabad'),
36 (102, 'Priya Sharma', '9876543211', 'Vijayawada'),
37 (103, 'Arjun Reddy', '9876543212', 'Bangalore'),
38 (104, 'Sneha Rao', '9876543213', 'Chennai'),
39 (105, 'Kiran Kumar', '9876543214', 'Hyderabad');
40
41 • INSERT INTO Account
42 (Account_No, Customer_ID, Account_Type, Balance, Branch)
43 VALUES
44 (10001, 101, 'Savings', 50000, 'Hyderabad'),
45 (10002, 102, 'Savings', 75000, 'Vijayawada'),
46 (10003, 103, 'Current', 120000, 'Bangalore'),
47 (10004, 104, 'Savings', 45000, 'Chennai'),
48 (10005, 105, 'Current', 90000, 'Hyderabad');
49
50 • INSERT INTO Bank_Transaction
51 (Account_No, Transaction_Type, Amount)
52 VALUES

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51 (Account_No, Transaction_Type, Amount)
52 VALUES
53 (10001, 'DEPOSIT', 10000),
54 (10002, 'DEPOSIT', 15000),
55 (10003, 'WITHDRAW', 20000),
56 (10004, 'DEPOSIT', 5000),
57 (10005, 'WITHDRAW', 10000);
58
59 • SELECT * FROM Customer;
60
61 • SELECT * FROM Account;
62
63 • SELECT * FROM Bank_Transaction;
64
65 • START TRANSACTION;
66
67 • UPDATE Account
68 SET Balance = Balance + 5000
69 WHERE Account_No = 10001;
70
71 • SELECT *
72 FROM Account
73 WHERE Account_No = 10001;
74
75 • COMMIT;
76
84 SET Balance = Balance - 10000
85 WHERE Account_No = 10001;
86
87 • SELECT *
88 FROM Account
89 WHERE Account_No = 10001;
90
91 • ROLLBACK;
92
93 • SELECT *
94 FROM Account
95 WHERE Account_No = 10001;
96
97 • START TRANSACTION;
98
99 • UPDATE Account
100 SET Balance = Balance + 5000
101 WHERE Account_No = 10001;
102
103 • SAVEPOINT Deposit1;
104
105 • UPDATE Account
106 SET Balance = Balance - 3000
107 WHERE Account_No = 10002;
108
109 • SAVEPOINT Withdrawal1;

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246 • START TRANSACTION;
247
248 • SELECT Balance
249 FROM Account
250 WHERE Account_No = 10001;
251
252 • COMMIT;
253
254 • SET SESSION TRANSACTION ISOLATION LEVEL SERIALIZABLE;
255
256 • START TRANSACTION;
257
258 • SELECT *
259 FROM Account
260 WHERE Account_No = 10001;
261
262 • COMMIT;
263
264 • START TRANSACTION;
265
266 • UPDATE Account
267 SET Balance = Balance + 5000
268 WHERE Account_No = 10001;
269
270 • SAVEPOINT AfterDebit;
271
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```
322 • UPDATE Account
323 SET Balance = Balance + 20000
324 WHERE Account_No = 10003;
325
326 • COMMIT;
327
328 • START TRANSACTION;
329
330 • UPDATE Account
331 SET Balance = Balance + 5000
332 WHERE Account_No = 10001;
333
334 • SAVEPOINT Deposit1;
335
336 • UPDATE Account
337 SET Balance = Balance - 2000
338 WHERE Account_No = 10002;
339
340 • ROLLBACK TO SAVEPOINT Deposit1;
341
342 • COMMIT;
343
344 • SELECT @@SESSION.transaction_isolation;
345
346 • SET SESSION TRANSACTION ISOLATION LEVEL SERIALIZABLE;
```

	Customer_ID	Customer_Name	Phone	City
▶	101	Hasini	9876543210	Hyderabad
	102	Priya Sharma	9876543211	Vijayawada
	103	Arjun Reddy	9876543212	Bangalore
	104	Sneha Rao	9876543213	Chennai
	105	Kiran Kumar	9876543214	Hyderabad
★	NULL	NULL	NULL	NULL

	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
▶	1	10001	DEPOSIT	10000.00	2026-08-30 15:40:03
	2	10002	DEPOSIT	15000.00	2026-08-30 15:40:03
	3	10003	WITHDRAW	20000.00	2026-08-30 15:40:03
	4	10004	DEPOSIT	5000.00	2026-08-30 15:40:03
	5	10005	WITHDRAW	10000.00	2026-08-30 15:40:03
	6	10001	DEPOSIT	10000.00	2026-08-30 15:44:22
	7	10002	DEPOSIT	15000.00	2026-08-30 15:44:22
	8	10003	WITHDRAW	20000.00	2026-08-30 15:44:22
	9	10004	DEPOSIT	5000.00	2026-08-30 15:44:22
	10	10005	WITHDRAW	10000.00	2026-08-30 15:44:22
★	NULL	NULL	NULL	NULL	NULL

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
	10002	102	Savings	111000.00	Vijayawada
	10003	103	Current	140000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
★	NULL	NULL	NULL	NULL	NULL